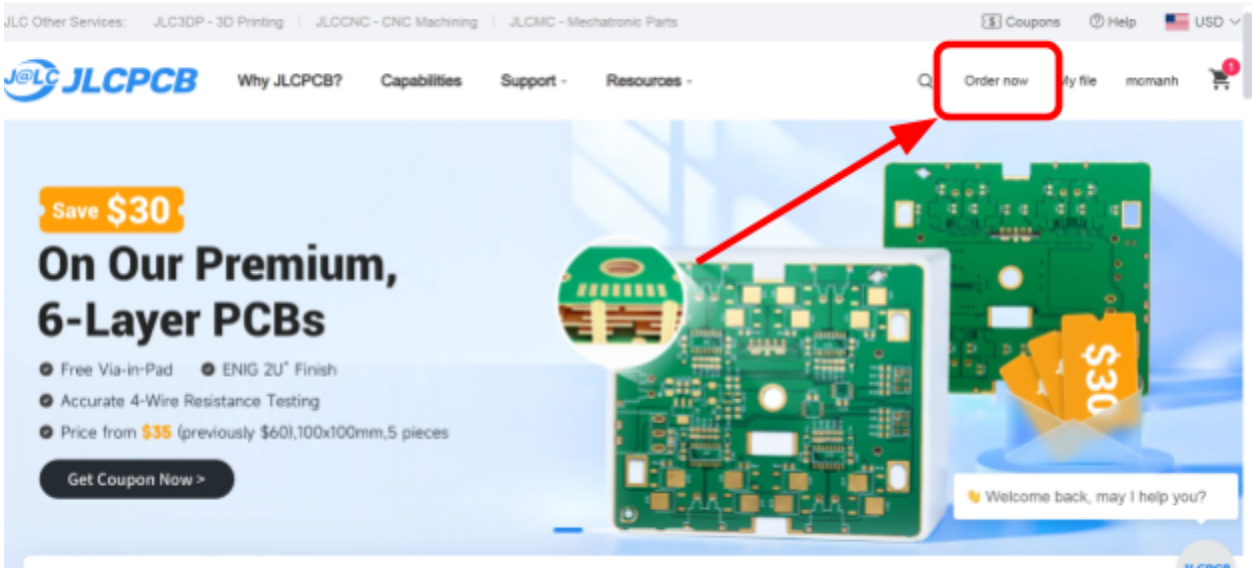
Step 1: Create an order

Navigate to JLCPCB and make a free account if you do not have one

Select “Order Now” in the top right of your screen



Step 2: Upload Gerber

Click “Add Gerber File” in the middle of the screen

Navigate to your Cross Point Project repository or clone it if you do not already have it, or alternatively follow the direct links in each step where a file is required

- Navigate to the zip file
  Cross-Point-Project→PCB Projects→cross-point-exp-brd-mini-red2-sjmp →Gerbers→cross-point-exp-brd-mini-red2-sjmp.zip

OR

- Download zip directly
  Follow this link
  Press “view raw” to download

Select and upload the zip file

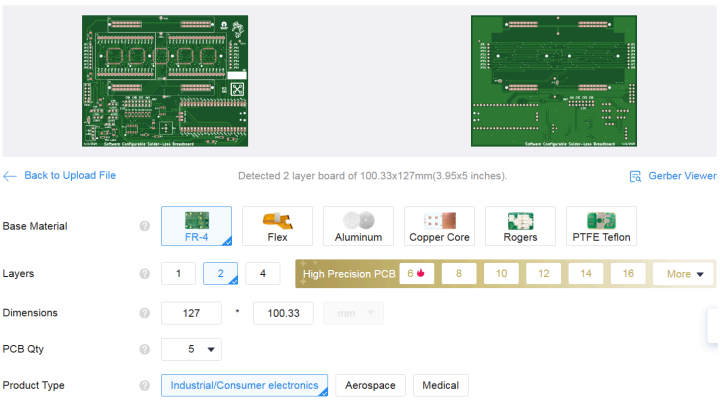
Step 3: Select Build Specifications

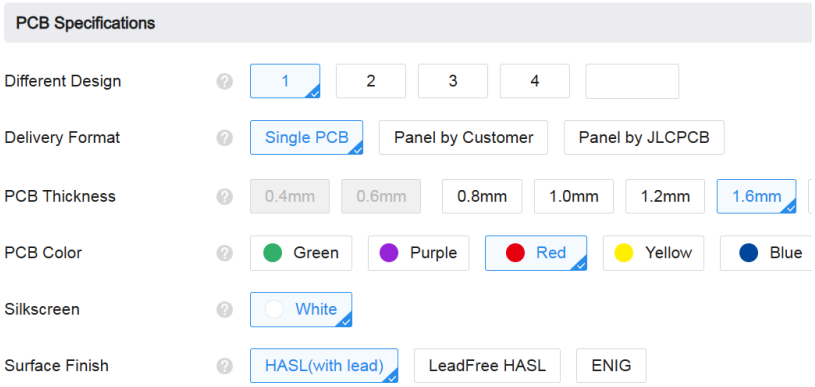
The specifications are the defaults from JLCPBC with the following changes:

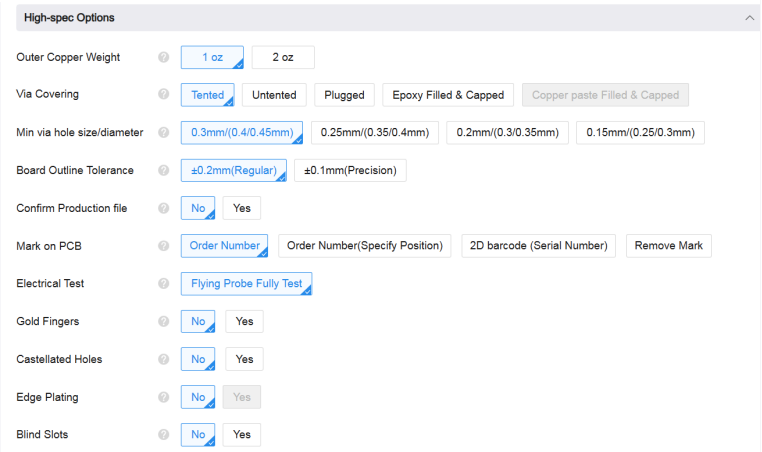
- PCB color **RED**
- PCB assembly **ON**
- PCBA type **STANDARD**
- Confirm Parts Placement **YES** (will have a pop up menu, check the box for “Do not confirm automatically”)

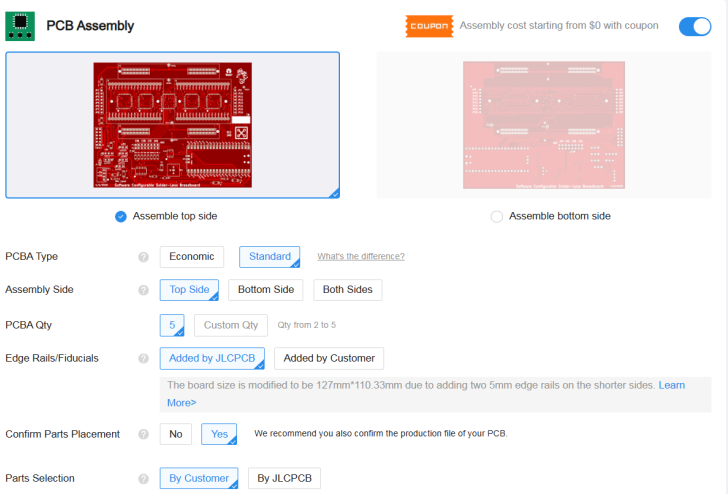
Price should be $9.40 at this point. Press “Next” below calculated price

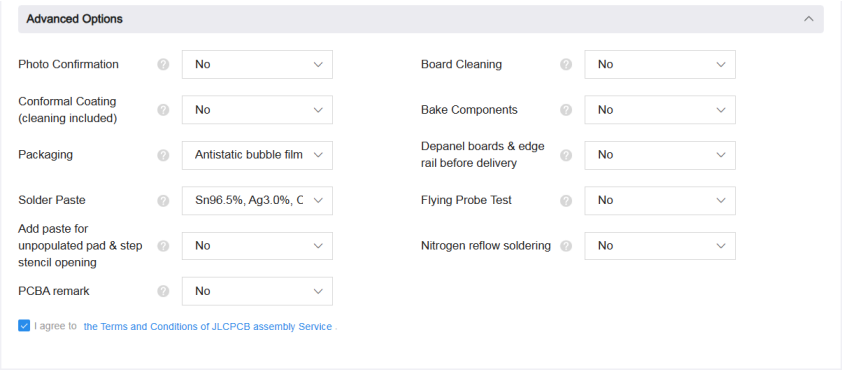
Specification images:





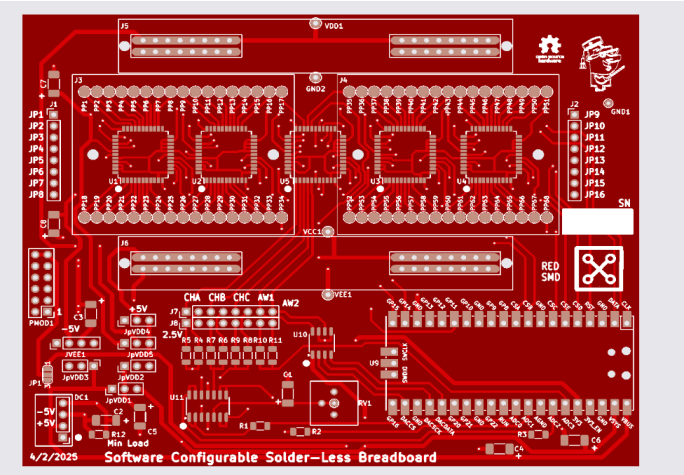




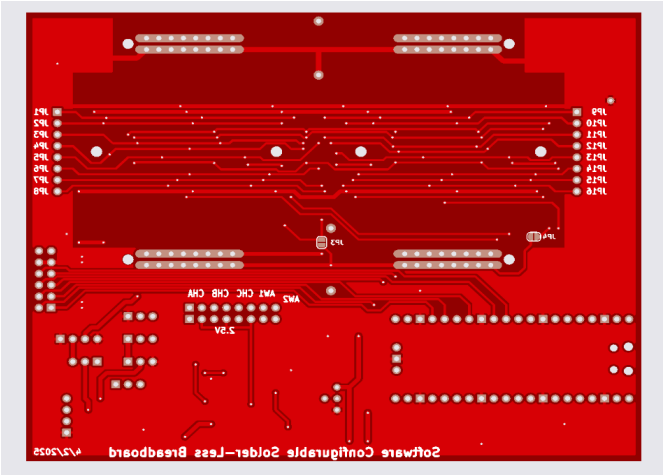


Step 4: Confirm Board Image

Top of Board



Bottom of Board



Confirm that your board matches with these images and select “NEXT”

Step 5: Upload BOM and CPL files

NOTE: It is possible that the sheets contain extra, blank pages. If there are any delete them as the upload will fail if these files have multiple sheets

BOM

Navigate to BOM:
Cross-Point-Project → PCB Projects → cross-point-exp-brd-mini-red2-sjmp → JLCPCB_BOM_RED2_SMD.xlsx

OR

Directly download BOM:
Follow this link and press “view raw” to download

CPL

Navigate to CPL:
Cross-Point-Project → PCB Projects → cross-point-exp-brd-mini-red2-sjmp → JLCPCB-red2-sjmp-top-pos.csv

OR

Directly download CPL:
Follow this link and press “download raw file

Click “Process BOM and CPL” which may take a few attempts to be accepted

Step 6:Confirm BOM

Check values to be consistent with the design you are implementing, and select all checkboxes on the right side of the screen.

Note that by default Pogo pins will be deselected to ensure you notice the additional charge for their placement on the board (at .08/per pogo or 11.52/board)

Uploaded BOM Data			Review Matched Parts					
Top Designator	Comment	Footprint	Matched Part Detail	Qty	Source	Lib Type	Total Cost	Select
C1,C3,C4,C5,C6	470nF	1206	GRM31C5C1E474JE01L C6049896 25V 470nF ±5% 1206 Multilayer Ceramic Capa...	25	JLCPCB	Extended	$11.5125	☑
C2,C7,C8	10uF	1206	CL31A106KOHNNNE C1849 16V 10uF X5R ±10% 1206 Multilayer Ceramic ...	23	JLCPCB	Extended	$0.5911	☑
DC1	RFM-0505S						$8.2575	☑
J1,J2,J7,J8	2.54-1*8PFe...						$2.7840	☑
PP1,PP2,PP3,PP...	BWCD-L3.5...	SMDPogoPinS...	BWCD-L3.5W2.0H1.0 C2826547 - Pogo Pin Spring Probe Connector ROHS	344	JLCPCB	Extended	$11.5240	☑
R1,R2,R3	220Ω	1206	1206W4F2200T5E C17957 250mW Thick Film Resistors 200V ±100ppm/℃...	25	JLCPCB	Extended	$0.0750	☑
R4,R6,R8	330KΩ	1206	1206W4F3303T5E C17929 250mW Thick Film Resistors 200V ±100ppm/℃...	25	JLCPCB	Extended	$0.0750	☑
R5,R7,R9	680KΩ	1206	1206W4F6803T5E C17974 250mW Thick Film Resistors 200V ±100ppm/℃...	25	JLCPCB	Extended	$0.0700	☑
R12	470Ω	1206_R	1206W4F4700T5E C4502 250mW Thick Film Resistors 200V ±100ppm/℃...	20	JLCPCB	Extended	$0.0580	☑
RV1	3386P-1-103...	Plugin-3P,9.5x9...	3386P-1-103TLF C116302 ±10% ±100ppm/℃ 500mW 10kΩ Plugin-3P,9.5...	5	JLCPCB	Extended	$12.1200	☑
U1,U2,U3,U4,U5	CH446Q	LQFP-44(10x10)	CH446Q C109471 LQFP-44(10x10) Analog Switches, Multiplexers...	25	JLCPCB	Extended	$33.8625	☑
U9	RapberryPiPico		PICO C7203002 - Raspberry Pi ROHS	5	JLCPCB	Extended	$32.3250	☑
U10	MCP4822-E/...	SOIC-8	MCP4822-E/SN C16040 4.5us SPI 2.7V~5.5V 12 Bits SOIC-8 Digital to ...	5	JLCPCB	Extended	$15.3975	
U11	MCP6004T-I/...	SOIC-14	MCP6004T-I/SL C7378 Four channels 1pA 0.6V/us 1MHz Rail-to-rail in...	5	JLCPCB	Extended	$2.4600	☑

The processing of this component is difficult. A special component fee of $0.0800 per piece will be charged.
*After placing an order, it will be reviewed. If PCBA advanced option is required, additional fees will be charged. The review result will be notified to you by email.

Welcome back, may I

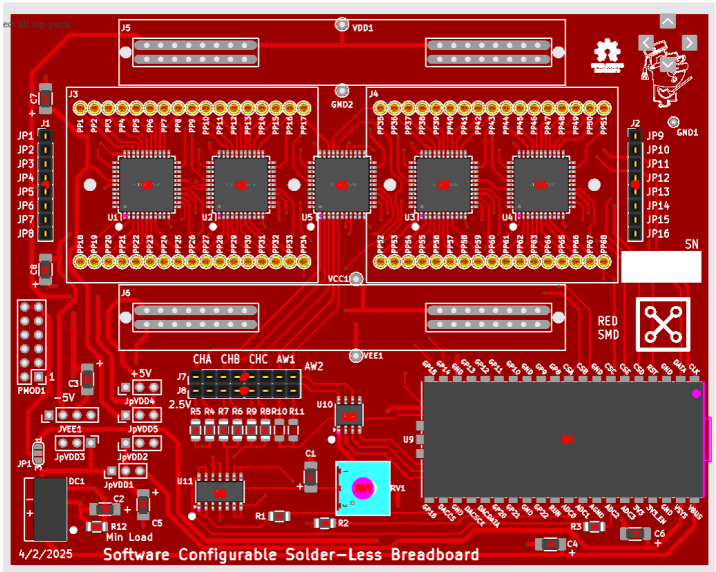
+ Add Part

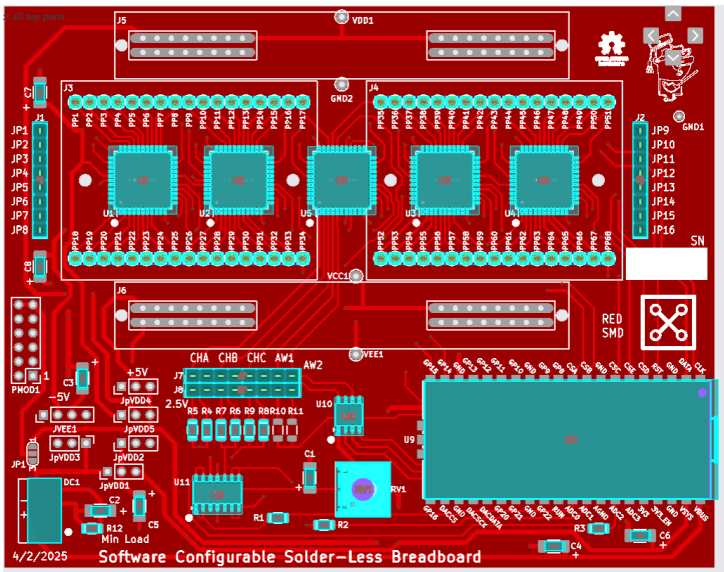
Please carefully check the packages of selected parts before proceeding.

Press “NEXT”

Step 7: Confirm Board

Confirm that your board matches this image with components placed and press “NEXT”



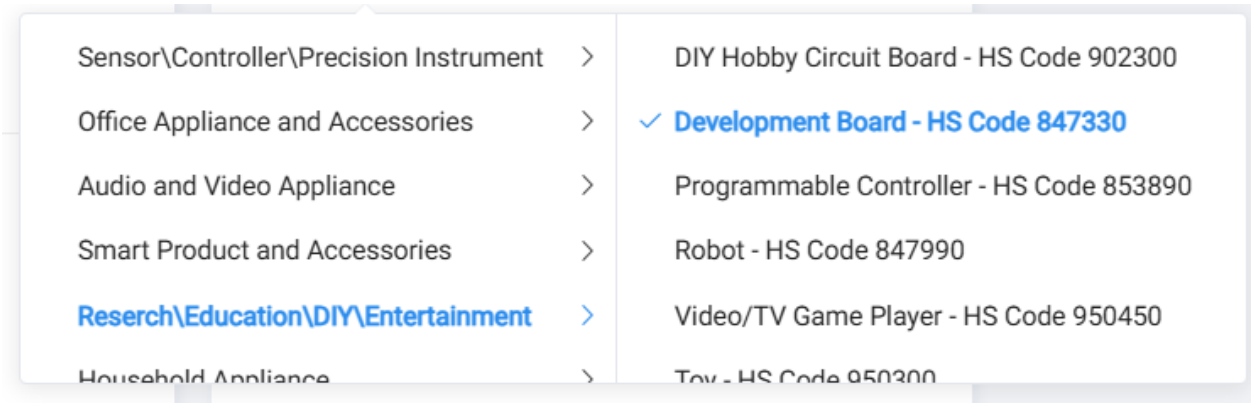


Step 9: Confirm Order

On the next screen you will see a bill for all materials and engineering for 5 boards, the total price in the top right should read $228.14, with a product weight of 1.47kg

Note that you can pay a surcharge for expedited assembly, which will increase price

To check out you must select a product description, this one seems like the most apt description, however I am unsure if this decision affects the order



Select “Save to cart” and follow checkout instructions to place order

Step 10: Final Assembly

Upon receiving boards from JLC, you will need to mount the breadboards to the pogo pins before use.

Per board, order four “Narrow Fillister Head Slotted Screw, 18-8 Stainless Steel, High-Profile, 2-56 Thread, 1/2" Long and four Low-Strength Steel Hex Nut Zinc-Plated, 2-56 Thread Size to attach two 170 Tie-Point Boards. Lastly you will need two power rails per board: you may choose between 50 pins and 60 pins, both conveniently sold in packs of two, so buy one pack per board.

To attach, simply place each breadboard upon the pogo pins, thread the screws through the board holes, PCB holes, and the hex nut and tighten. The power rails can be stuck directly to the board by using the sticker on the back of the rails. A wire must lastly be soldered from the back of the board to the power rails, connecting to the underside of J5 and J6. It is recommended to test each column of the breadboard with a DMM in continuity mode.